Fiscal Year:	FY 2005	**Task Last Updated:**	FY 04/26/2006
PI Name:	Gast, Alice P. Ph.D.		
Project Title:	Anisotropic Colloidal Self Assembly		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Complex fluids		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	02139	**Congressional District:**	8
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	98-HEDS-03
Start Date:	03/01/2000	**End Date:**	11/01/2004
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	1
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
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Flight Program:	ISS		
Flight Assignment:	ISS		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NAG3-2832		
Performance Goal No.:	1.23		
Performance Goal Text:	Pursue ground-breaking basic research in fluid physics and transport phenomena to provide fundamental understanding of natural phenomena affected by gravity for increased efficiency and effectiveness of space-based and industrial processes.		

Task Description:	Magnetorheological (MR) fluids, suspensions of paramagnetic particles in a nonmagnetic fluid, are a class of tunable materials that can change from liquid to solid upon the application of a magnetic field. The rheological properties of these materials are determined by their microstructure, which transitions from isolated chains to a percolated network at some critical particle concentration. However, their microstructure is difficult to study due to sedimentation effects which become increasingly important as the particles aggregate into larger structures. Understanding MR fluids near and around this gelation point is important in designing “smart fluids” for various applications such as shock absorbers, robotic arms or astronaut-supporting suits. In the past year, we focused primarily on ground-based experiments which address some of the fundamental questions surrounding the structure and dynamics of chains near the threshold for formation of a fully cross-linked network. While several studies have aimed to describe the rheological properties of MR fluids on a microstructural basis, we seek to investigate the fundamental micromechanics of these structures to understand their rheological behavior from a microscopic level. We also seek to study phase transitions in reduced dimensions caused by magnetic interactions. These experiments were originally intended to take place on the ISS in the CHAINS project, which was cancelled in May 2004. In order to achieve the goals we set out in the CHAINS project, we turned to high-speed video experiments. These experiments allow us to study chain structure and dynamics when the networks first form, thereby circumventing the issue of sedimentation. We use a high speed camera with a frame capture of 2000 frames/second, in conjunction with either microscopy or diffusive wave spectroscopy (DWS). In MR suspensions, dipolar particles aggregate to form long chains. More concentrated systems exhibit a rapid liquid to solid transition as chains crosslink, and are capable of supporting stresses perpendicular to the applied field. These unique properties have many possible technological applications, especially for actuator-type devices. These experiments should elucidate the nature of the gelation transition in MR fluids, and provide insight into the rheological properties of these tunable materials.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	In order to study aggregation without the effects of sedimentation, we use a high-speed camera to image the microstructure and scattering of MR fluids. With these extremely fast images we can observe the transition from an isotropic suspension to a cross-linked structure at timescales shorter than the sedimentation time. We studied the gelation transition with microscopic images at 10 and 50X magnification over these short time scales (less than 2 seconds). The frame capture rate for our experiments was 2000 frames/second. We used image processing techniques such as Fourier transforms, crosscorrelation and thresholding to characterize the structure growth and dynamics. In a complementary study, we have built a diffusing wave spectroscopy (DWS) setup to measure the dynamics as we cross the threshold volume fraction for gelation. DWS is ideal for measuring the fluctuations of dense, strongly scattering samples such as these. We use a camera to image the forward scattering, with each pixel acting as an independent detector. This allows us to average over pixels instead of time-averaging; thus, we can obtain excellent statistics in a short amount of time. We follow the scheme outlined in [1] for multi-speckle DWS. The sample is illuminated by an expanded laser beam. The scattered light passes through an aperture, then is collected by a camera. Images are saved directly to a computer. We use an IR laser (lambda = 894) in order to minimize absorption. We have now optimized our DWS setup, and are prepared to measure time intensity correlation functions for different particle volume fractions. [1] V. Viasnoff, F. Lequeux, and D. Pine, Rev. Sci. Intrum. 73, 2336 (2002).
Bibliography Type:	Description: (Last Updated: 04/19/2021)
Articles in Peer-reviewed Journals	Biswal SL, Gast AP. "Micromixing with linked chains of paramagnetic particles." Anal Chem. 2004 Nov 1;76(21):6448-55. https://doi.org/10.1021/ac0494580 ; PMID: 15516140 , Nov-2004
Articles in Peer-reviewed Journals	Biswal SL, Gast AP. "Rotational dynamics of semiflexible paramagnetic particle chains." Phys Rev E Stat Nonlin Soft Matter Phys. 2004 Apr;69(4 Pt 1):041406. https://doi.org/10.1103/PhysRevE.69.041406 ; PMID: 15169018 , Apr-2004
Articles in Peer-reviewed Journals	Lyles BF, Terrot MS, Hammond PT, Gast AP. "Directed patterned adsorption of magnetic beads on polyelectrolyte multilayers on glass." Langmuir. 2004 Apr 13;20(8):3028-31. https://doi.org/10.1021/la049486d ; PMID: 15875825 , Apr-2004
Presentation	Bentley, K.; Gast, A.P. "Magneto-rheological materials in space" AIChE Annual Meeting Oct-2004